

Causality

Various Methods and Their Characteristics (Part III)

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Roadmap

Causal Inference is not just about adjustments! Various methods are applicable with different scenarios.

- Matching
- Instrumental Variables
- Differences-in-Differences
- Regression Discontinuity
- Synthetic Control Method

Overview

Matching

Instrumental Variable

Regression Discontinuity

Differences-in-Differences

Synthetic Control Method

Summary

Matching: Motivation

- Causal inference $\rightsquigarrow$ inference for counterfactuals through comparison between treated and control units
- Consider the Average Treatment Effect **for the Treated** (ATT):

$$\tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid X = 1]$$

- Regression $\rightsquigarrow$ model-based imputation (N_1 , # of the treated):

$$\hat{\tau}_{\text{reg}} = \frac{1}{N_1} \sum_{i=1}^N X_i (Y_i - \hat{\mu}_0(\mathbf{z}_i))$$

- Matching $\rightsquigarrow$ nonparametric imputation.
- Matching can help identify control units that are comparable to treated units *before* outcomes are observed.
 $\implies$ Useful in settings where we must decide in advance which control individuals to follow up for outcome measurement.

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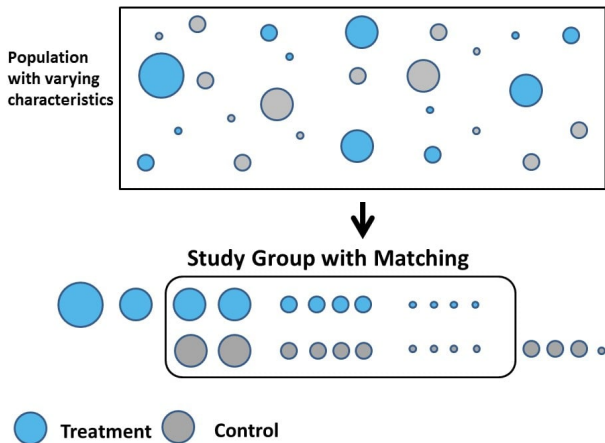
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Case Study: Matching in the NSW Job Training Program

- **Context:** The National Supported Work (NSW) Demonstration was a randomized job training program 🏠 conducted in the 1970s targeting disadvantaged groups 👤.
- **Research Question:** What is the causal effect of job training on post-program earnings?
- **Method:** Use **matching** to estimate the treatment effect by comparing NSW participants to similar individuals from observational datasets.
- **Insight:** Highlights how matching on observed covariates can recover experimental estimates.

Matching in a Figure

A large pool of potential control units would enable each treated unit to be matched to one or more distinct controls, even without replacement.



Discussion

- Matching still assumes *unconfoundedness*.
- Let $\mathcal{M}_i$ be the “matched set” for treated unit i , that is, the units in $\mathcal{M}_i$ share similar characteristics with unit i . Then,

$$\hat{\tau}_{\text{match}} = \frac{1}{N_1} \sum_{i=1}^N X_i \left(Y_i - \underbrace{\frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} Y_j}_{Y_i(0)} \right)$$

- Researchers must decide on matching ratios (one-to-one vs. one-to-many), whether to allow replacement, how to define similarity (e.g., distance metrics, propensity scores), and how to weight matched units.
- Methods include exact matching, coarsened exact matching, nearest neighbor matching, or optimal matching.

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Instrumental Variable

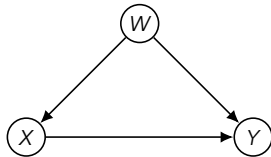
Regression Discontinuity

Differences-in-Differences

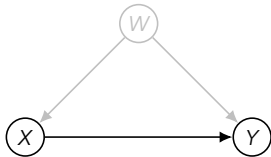
Synthetic Control Method

Summary

Instrumental Variable: Motivation



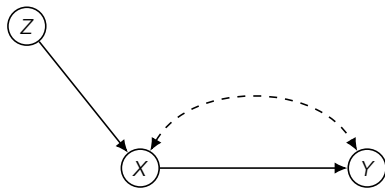
Instrumental Variable: Motivation



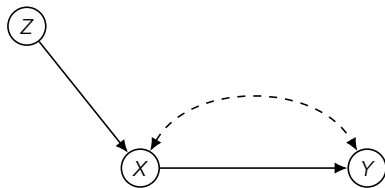
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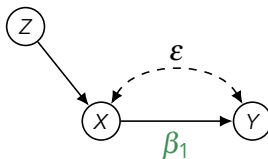
Instrumental Variable: Motivation



Z is an **instrument variable** (IV):

- Relevance: The instrument must affect the treatment X.
- Exclusion Restriction: The instrument must not directly affect outcome Y; it should influence the outcome only through the treatment X.

IV under Linear Model

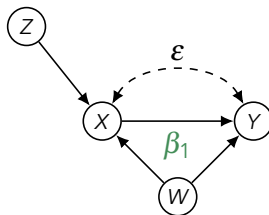


- Traditional linear model of Y being related to a scalar treatment X

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where the estimand is the coefficient β_1

IV under Linear Model

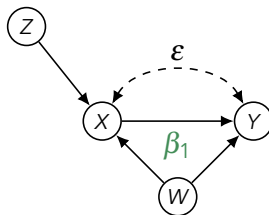


- Traditional linear model of Y being related to a scalar treatment X given a set of covariates W

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- Challenge:** the error term ε is dependent of treatment variable X (confounded). $\implies$ Direct OLS estimator of β_1 is **biased**

Two-Stage Least Squares (2SLS) with Covariates

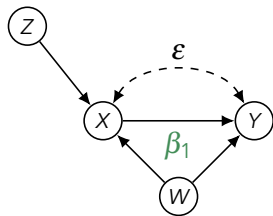
- Goal ©: Estimate the causal effect of treatment X on outcome Y , controlling for covariates W , in the presence of endogeneity.
- Stage 1: Regress X on instrument Z and covariates W

$$X = \pi_0 + \pi_1 Z + \pi_2^\top W + \varepsilon$$

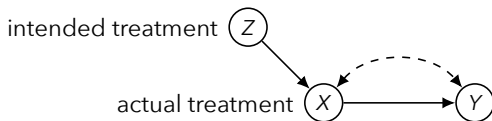
- Stage 2: Regress Y on the fitted values $\hat{X}$ from stage 1 and covariates W

$$Y = \beta_0 + \beta_1 \hat{X} + \beta_2^\top W + \varepsilon$$

- *Intuition:* Use only variation in X that is explained by the instrument Z , while adjusting for W , i.e., recover the “unconfounded portion”



Noncompliance and LATE



- Z is randomly assigned, but actual treatment X may differ due to **noncompliance**.
- IV can still identify causal effects in such cases by focusing on the subset of individuals who **comply** with their assignment.
 $\implies$ **Compliers** are units for which $X(Z=0) = 0$ and $X(Z=1) = 1$ (**unknown!**)
- The **Local Average Treatment Effect (LATE)** refers to the causal effect for these **compliers**.
- LATE may differ from the average treatment effect (ATE) in the population, but it remains a valid and interpretable causal quantity under certain assumptions.

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Regression Discontinuity Design (RDD)

- RDD is a quasi-experimental design—an approach that approximates randomized experiments using non-randomized data.
- Treatment assignment changes discontinuously based on a pre-treatment variable (the **running** or **forcing** variable).
- A fixed **cutoff** ✂ determines treatment, while unobserved confounders are assumed to vary smoothly around the cutoff.
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🔑: knowing the assignment rule enables causal inference
- Basic idea: in these studies, comparing **units with similar values** of this variable, but different levels of treatment would lead to causal effect of the treatment **at the threshold**
- The discontinuity is often created by a pre-fixed, artificial threshold of a pre-treatment variable

Examples

Example 1 Scholarships based on test score cutoff (Thistlethwaite and Campbell, 1960)

- Eligibility depends solely on whether a student's **test score** is above or below a threshold
- Students near the threshold are arguably **comparable** in background (e.g., ability, attitude)

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Example 2 Party victory in close elections (Lee, Moretti, and Butler, 2004)

- Whether a Democrat barely wins or loses a House election depends on the **vote share** crossing 50%
- Districts near the 50% cutoff are arguably **comparable** in voter preferences
- Used to study causal effect on legislators' **ideological positions** (e.g., ADA scores)

RD in a Figure

Find a cutpoint c , which determines the treatment assignment s.t. $X_i = 1 \{Z_i \geq c\}$

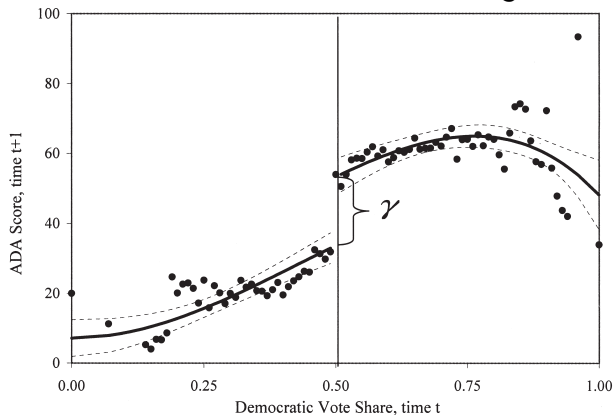


FIGURE I
Total Effect of Initial Win on Future ADA Scores: γ

- Quantity of interest: Local ATE at the cutoff c

$$\begin{aligned}\tau_{RD} &= \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c] \\ &= \mathbb{E}[Y_i(1) \mid Z_i = c] - \mathbb{E}[Y_i(0) \mid Z_i = c]\end{aligned}$$

- Typically, we never observe any $Z_i = c$ (continuous Z) and Positivity is violated

$$\mathbb{P}[X_i = 1 \mid Z_i = c - \varepsilon] = 0 \quad \text{and} \quad \mathbb{P}[X_i = 1 \mid Z_i = c + \varepsilon] = 1$$

- 🔑 Identification comes from extrapolation around c , which requires assumption (e.g., smoothness).
- Estimate the treatment effect by fitting local linear regressions on both sides of the cutoff, typically using a kernel-weighted approach.

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Case Study: Minimum Wages (Card & Krueger, 1994)

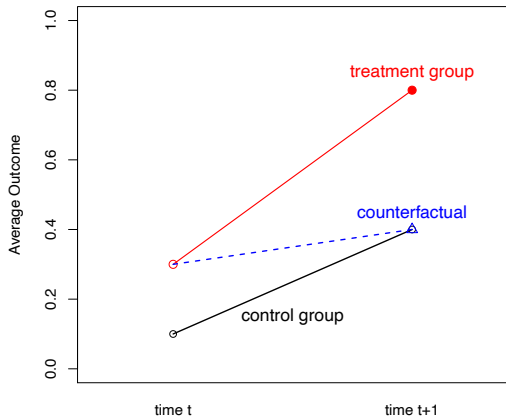
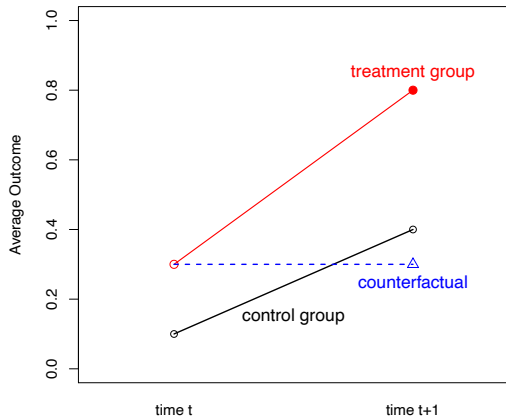
Q: Does increasing the minimum wage affect employment?

- Classical economic theory tends to point to negative effects.
But difficult to randomize changes to the minimum wage.
- In 1992, **NJ** minimum wage increased from \$4.25 to \$5.05
 - ▶ Neighboring **PA** stays at \$4.25
 - ▶ We observe employment in both states before and after increase
- Look at **eastern PA** and **NJ** fast food restaurants.
 - ▶ Similar prices, wages, products, etc.
 - ▶ Most likely to be affected by the change.

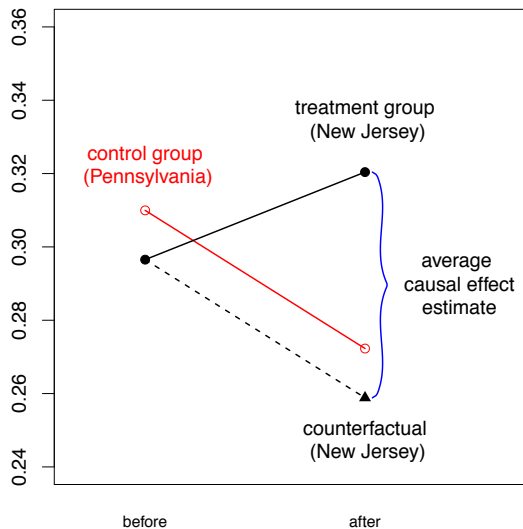
Variations to Exploit

There are two possible sources of variation to utilize:

- Exploit variation in treatment **within a unit over time** (before/after)
- Exploit **cross-sectional** variation in treatment.



Differences in Differences in a Figure



Differences in Differences Design

- Basic setup: two **groups** $G_i \in \{0, 1\}$, two time **periods** $t \in \{0, 1\}$.
- Treatment status in each period:
 - ▶ In control group, $G_i = 0 \rightsquigarrow X_{i1} = 0$,
 - In treated group, $G_i = 1 \rightsquigarrow X_{i1} = 1$,

	Time period	
	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$X_{i0} = 0$	$X_{i1} = 1$
Control group ($G_i = 0$)	$X_{i0} = 0$	$X_{i1} = 0$

- $Y_{it}(x)$ is the potential outcome under treatment x at time t .
Can you draw observed potential outcomes on the previous plot? 😊

Identification problem and Parallel Trends

- Average treatment effect on the treated (ATT):

$$\begin{aligned}\tau_{\text{ATT}} &= \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] \\ &= \mathbb{E}[Y_{i1}(1) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) \mid G_i = 1] \\ &= \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{\text{given}} - \underbrace{\mathbb{E}[Y_{i1}(0) \mid G_i = 1]}_{\text{unknown}}\end{aligned}$$

- Key assumption of DiD: parallel trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$$

- **Interpretation:**

a trend in the control group is a good proxy how the treated group would have changed over time without treatment.

DiD: Detail

$$\tau_{\text{ATT}} = \mathbb{E}[Y_{i1}(1) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) \mid G_i = 1]$$

	Time period	
	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$\mathbb{E}[Y_{i0}(0) \mid G_i = 1]$	$\mathbb{E}[Y_{i1}(1) \mid G_i = 1]$
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Difference-in-differences:

- Assumption: parallel trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$$

- Use pre-period treated outcome plus trend in control group:

$$\tau_{\text{ATT}} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0])$$

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Control group ($G_i = 0$)	$\mathbb{E}[Y_{i0}(0) \mid G_i = 0]$	$\mathbb{E}[Y_{i1}(0) \mid G_i = 0]$

Difference-in-differences:

- Assumption: parallel trends

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] + \mathbb{E}[Y_{i0}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1}(0) \mid G_i = 1]$$

- Use pre-period treated outcome plus trend in control group:

$$\tau_{\text{ATT}} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0])$$

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Synthetic Control Method

- Designed for settings with **a single treated unit** and many potential controls.
 - ▶ Compare the **outcome trajectory** of the treated unit to that of a control group.
 - ▶ But which controls should we compare to? Not all are comparable.
 - ▶ Choosing the wrong comparison group can lead to biased estimates.
- Particularly useful when there is **no natural control group** or when the **parallel trends assumption is questionable**.
- **Synthetic control**: constructs a weighted average of control units to best replicate the pre-treatment outcomes of the treated unit.
The weights are also chosen to minimize differences in pre-treatment characteristics.

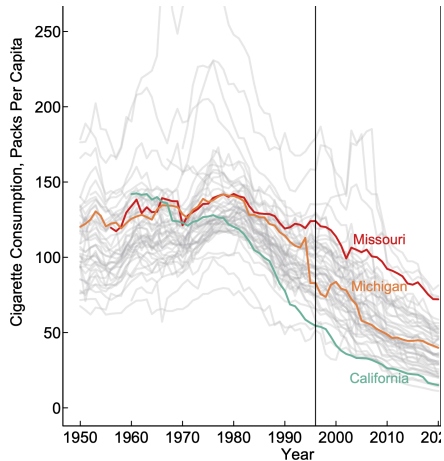
Case Study: Proposition 99 in California (Abadie et al., 2010)

Q: Did California's 1989 tobacco control program reduce cigarette consumption?

- In 1988, California passed **Proposition 99**:
 - ▶ Introduced a substantial cigarette tax increase.
 - ▶ Funded anti-smoking campaigns and public health programs.
- Standard DiD approach is hard:
 - ▶ No obvious single control state; Pre-treatment trends differ across states.
- **Synthetic Control Method (SCM)**:
 - ▶ Build a weighted combination of other states to construct a "synthetic California" under control.
 - ▶ Match on cigarette sales (Y) and covariates in **pre-1989** period.
- Compare **post-1989** smoking trends.

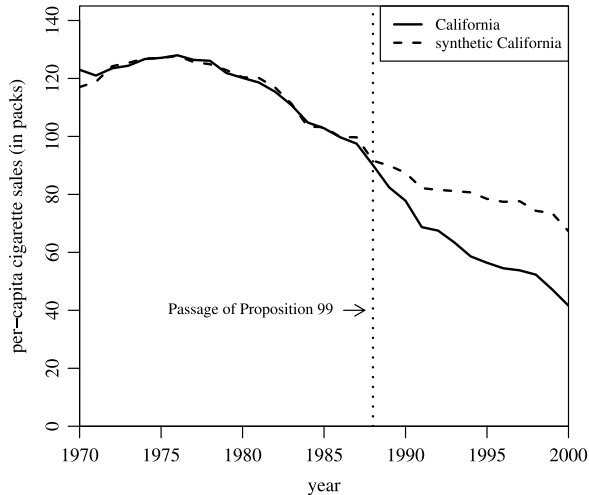
Synthetic Control in a Figure

Let's examine the data from California and other states.



Synthetic Control in a Figure

States are selected sparsely: Colorado, Connecticut, Montana, Nevada, Utah



Synthetic Control

- By consistency, for $t > T_0$ (time of treatment):

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0) = Y_{1t} - Y_{1t}(0)$$

- **Synthetic control**: Choose weights $\mathbf{w}$ such that:
 - ▶ each weight is ≥ 0 and weights are summed to 1 (i.e., convex)
 - ▶ for the **pre-treatment** period, $t \leq T_0$, minimize

$$\left| Y_{1t} - \sum_{j \in \text{candidates}} w_j Y_{jt} \right|, \quad \left| \mathbf{z}_1 - \sum_{j \in \text{candidates}} w_j \mathbf{z}_j \right|$$

- We hope, for the **post-treatment** period: $\sum_{j \in \text{candidates}} w_j Y_{jt} \approx Y_{1t}(0)$

Discussion

- Synthetic control methods offer practical benefits for estimating the impact of policy interventions and major events.
- As with any causal inference method, the validity of results hinges on careful implementation and on meeting the necessary data and contextual assumptions.
- Many policies of interest to economists operate at the **aggregate level**, targeting entire units such as states, regions, or countries.

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- **Matching:** Compares treated and control units with similar observed covariates.
- **Instrumental Variables:** Uses external variation to address endogeneity in treatment.
- **Differences-in-Differences:** Compares outcome changes over time between groups.
- **Regression Discontinuity:** Exploits cutoff-based assignment for local causal inference.
- **Synthetic Control Method:** Constructs a weighted combination of control units to estimate counterfactuals.